

Antiemetic activity of volatile oil from *Mentha spicata* and *Mentha × piperita* in chemotherapy-induced nausea and vomiting

Background: This study is aimed at determining the efficacy of *Mentha spicata* (*M. spicata*) and *Mentha × piperita* (*M. × piperita*) in preventing chemotherapy-induced nausea and vomiting (CINV). **Methods:** This was a randomised, double-blind clinical trial study. Prior to the study, patients were randomly assigned into four groups to receive *M. spicata* or *M. × piperita*. Statistical analysis included the χ^2 test, relative risk, and Student's t-test. Fifty courses were analysed for each group that met our eligibility criteria. The treatment and placebo groups applied essential oils of *M. spicata*, *M. × piperita*, or a placebo, while the control group continued with their previous antiemetic regimen. Patients or guardians recorded the number of emetic events, the intensity of nausea over 20 h of chemotherapy, as well as any possible adverse effects that occurred during this time. **Results:** There was a significant reduction in the intensity and number of emetic events in the first 24 h with *M. spicata* and *M. × piperita* in both treatment groups ($p < 0.05$) when compared with the control and no adverse effects were reported. The cost of treatment was also reduced when essential oils were used. **Conclusion:** *M. spicata* or *M. × piperita* essential oils are safe and effective for antiemetic treatment in patients, as well as being cost effective.